

Combustion, Venting and Furnace Safety

Anything that burns fuel needs air in and exhaust out. Your six-times miss is here, and so is the reason behind it.

YOU HAVE MISSED THIS 6 TIMES

When the furnace is located in a small, enclosed area:

Inlet and outlet ventilation openings must be provided.

Six times missed, and the pull is to answer with one opening. It has to be two. Combustion consumes oxygen and produces hot exhaust, so a sealed closet would starve the burner and then spill its exhaust into the room. Low inlet for cool combustion air coming in, high outlet for warm air leaving — air cannot flow through a space with only one hole in it.

YOU HAVE MISSED THIS

Gas furnaces located in garages or basements need:

Outside combustion air.

The numbers here

Unconfined space — combustion air	50 cu ft per 1,000 BTU/h
Combustion air taken directly from outdoors	1 sq in per 4,000 BTU/h
Minimum working space in front of a gas furnace at the access panel	30 in
Gas shutoff valve — within this distance of the appliance	6 ft

The furnace working space is a genuine 30-inch depth

Do not carry it across to the electrical panel, where 30 inches is the WIDTH and the depth is 36. Same number, two different measurements on two different appliances.

Venting

The numbers here

Gas vent above an air intake — vertical	3 ft
Gas vent from an air intake — horizontal	10 ft
Type B vent — terminate at least this far from a vertical sidewall	8 ft
Type B vent — or at least this far above any obstruction within 10 ft	2 ft

YOU HAVE MISSED THIS 3 TIMES

A Type B vent should terminate at least ____ feet from a vertical sidewall, or at least two feet above any obstruction within 10 feet.

Eight.

Exhaust must not be drawn straight back into the building. Three feet above an intake or ten feet away from it horizontally — vertical separation works harder because exhaust rises.

Asked which statement about horizontal sections of smoke pipe is FALSE, the false one is that the pipe should be installed in a level position. It must rise toward the chimney, because hot flue gases rise and a level or falling run lets them stall and spill. On a FALSE question you are being asked to pick the statement that is untrue, so read every option as a claim to be checked rather than an answer to be recognized.

Reading combustion faults

Soot in and around the furnace	Indicates the vent pipe is clogged. Exhaust that cannot leave deposits its carbon where you can see it.
Defective thermocouple	The pilot light will not light. The thermocouple proves a flame is present; with it faulty, the gas valve refuses to stay open.
Puff back	A problem found in oil-fired systems — unburned fuel accumulating and then igniting all at once.

YOU HAVE MISSED THIS 2 TIMES

A "puff back" is a problem found in what type of heating system?

Oil-fired systems.

YOU HAVE MISSED THIS 2 TIMES

An effective and safe way of cutting fuel consumption in a furnace can be accomplished by all of the following EXCEPT:

Installing an automatic flue damper.

On an EXCEPT question, find the one that is not safe or not effective. A flue damper interferes with the exhaust path, which is a safety system — so whatever it saves is not worth it, and it is the odd one out.

Say it back

Cover the page. Answer each one out loud.

1. A furnace in a small enclosed area needs what, and why two rather than one?
2. Combustion air for an unconfined space, and for air taken from outdoors?
3. Working space in front of a gas furnace, and how it differs from the panel rule?
4. Gas vent clearances above and horizontally from an air intake?
5. Should a horizontal smoke pipe be level? Say why.
6. Soot around the furnace tells you what?
7. A defective thermocouple produces which symptom?
8. Which fuel system gets a puff back?

